

When “I’m Fine” Isn’t Fine: Response Strategy Selection Under Acoustic-Semantic Emotional Incongruence in Supportive Dialogue

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Abstract

Effective emotional support conversation (ESC) requires understanding not only what a speaker says, but how they say it. When acoustic expression contradicts the literal sentiment of words—*acoustic-semantic emotional incongruence*—text-only systems systematically misread the user’s emotional state. We investigate this challenge using MELD, identifying 232 utterances exhibiting *masking* (positive words, negative tone) and *overstatement* (negative words, positive tone). Four prompting baselines on Qwen2.5-7B-Instruct confirm that explicit mismatch acknowledgment substantially raises acknowledgment (0.22→0.74) and empathy (0.018→0.032), replicating the core trend of Tahir et al. (2025), yet treat all incongruence uniformly. Strategy-guided LoRA fine-tuning on ESConv across six strategies reveals that *Providing Suggestions* maximizes empathy on both subtypes, but strategies diverge sharply by type: *Restatement or Paraphrasing* reaches Appropriateness 0.659 on overstatement yet only 0.257 on masking, demonstrating that incongruence type is a critical determinant of strategy effectiveness. We contribute a reproducible five-metric evaluation suite and a controlled within-experiment baseline design for multi-modal ESC research.

1 Introduction

Emotional support conversation (ESC) systems aim to provide empathic, contextually appropriate responses to users in emotional distress (Liu et al., 2021). Most existing approaches rely exclusively on textual input to infer emotional state—an assumption that breaks down when a user’s acoustic expression contradicts their words. A speaker may say “I’m fine” in a trembling voice (*masking*), or complain dramatically with a lighthearted tone (*overstatement*); the true emotional state lies in the relationship between what is said and how it is said.

Tahir et al. (2025) establish that incongruence awareness improves empathic response quality, but

treat incongruence as a single undifferentiated phenomenon. A fundamental question remains: do *masking* and *overstatement*—two subtypes with opposite communicative functions—require qualitatively different empathic strategies? Because E-THER’s data, weights, and evaluators are not released, we replicate its trend through prompting on Qwen2.5-7B-Instruct over MELD’s incongruent subset (Poria et al., 2019), then ask the subtype-specific question by fine-tuning the same model on ESConv (Liu et al., 2021) across six strategies. Because fine-tuning itself reduces the model’s tendency to surface mismatches, the external prompting baseline is no longer directly comparable; we therefore introduce a within-experiment *No-Strategy* baseline that fixes model and prompt structure, isolating the effect of strategy selection alone.

Across 1,624 generations on 232 incongruent samples evaluated with five automatic metrics, strategy-guided fine-tuning outperforms No-Strategy on both subtypes. *Providing Suggestions* maximizes empathy across both, but strategies diverge sharply by type: *Restatement or Paraphrasing* achieves Appropriateness 0.659 on overstatement vs. 0.257 on masking, while *Reflection of Feelings* attains the highest emotion correctness on masking. Overstatement is consistently more tractable than masking.

This work contributes (i) a replicated mismatch-awareness trend on open models and naturalistic data, (ii) a five-metric evaluation suite and within-experiment baseline design, and (iii) the empirical finding that incongruence type is a critical determinant of strategy effectiveness—arguing for a shift from uniform-strategy to type-sensitive ESC design.

2 Related Work

Empathic dialogue research has evolved along two largely parallel tracks—multimodal emotion understanding and strategy-guided response generation—that have not been integrated in the context of acoustic-semantic incongruence; we situate our work at this intersection.

Multimodal empathy and incongruence-aware systems. Tahir et al. (2025) introduce E-THER, showing that incongruence-aware fine-tuning improves empathy over GPT-4V; however, it treats incongruence as a single phenomenon and is not publicly released, limiting both subtype analysis and reproducibility. Zhu et al. (2023) annotate empathy in counseling videos but focus on detection rather than generation; Zadeh et al. (2018) show audio and text carry complementary emotional information but rarely extend to response generation.

Emotional support conversation. Liu et al. (2021) introduce ESConv with eight Helping Skills strategies (Hill, 2009), establishing strategy-guided generation as a key ESC paradigm. Rashkin et al. (2019) contribute EmpatheticDialogues as a standard empathic-response benchmark, and subsequent work explores strategy planning (Tu et al., 2022) and reinforcement learning for strategy selection. All of these approaches operate on textual input alone, leaving a systematic gap when acoustic expression contradicts text.

Multimodal emotion in dialogue. MELD (Poria et al., 2019) provides per-utterance emotion labels across audio, video, and text from *Friends*; its naturalistic expressions—distress, sarcasm, social masking—make it well-suited for identifying acoustic-semantic incongruence beyond clinical domains. Recent LLM-based emotion work (Lei et al., 2023; Wu et al., 2025) demonstrates generative-model potential but does not address cross-modal incongruence.

LLMs and parameter-efficient adaptation. Instruction-tuned LLMs such as Qwen2.5 (Team, 2025) are well-suited for strategy-conditioned generation, and LoRA (Hu et al., 2022) enables low-cost domain adaptation. The interaction between explicit strategy guidance and incongruence-type awareness remains unexplored—a gap our work directly addresses.

3 Data

3.1 MELD

We use the Multimodal EmotionLines Dataset (MELD) (Poria et al., 2019), a multimodal conversational dataset derived from the TV series *Friends*. MELD contains 13,708 utterances across 1,433 dialogues, annotated with one of seven emotion labels (anger, disgust, fear, joy, neutral, sadness, surprise) per utterance, along with aligned audio and video recordings. Its naturalistic conversational setting—spanning humor, sarcasm, distress, and social performance—makes it well-suited for studying acoustic-semantic incongruence in everyday dialogue, in contrast to clinical datasets such as E-THER (Tahir et al., 2025).

3.2 Mismatch Identification

MELD provides per-utterance audio emotion labels but no text-side emotion annotations. To obtain text sentiment, we apply the Emotion English DistilRoBERTa-base classifier (Hartmann, 2022) to each utterance transcript. Both audio and text emotion labels are then collapsed to binary sentiment polarity: *negative* (anger, disgust, fear, sadness) or *positive* (joy). Utterances labeled as neutral or surprise are excluded, as their polarity is ambiguous.

Utterances where audio and text polarities conflict are retained and categorized into two subtypes: **masking** (positive text, negative audio) and **overstatement** (negative text, positive audio). Although the text classifier introduces some label noise, this does not undermine our analysis—our goal is to measure response quality trends across conditions rather than evaluate detection accuracy, so minor mislabeling acts as noise rather than systematic bias.

To partially balance the two subtypes, we supplement the 36 test-split masking samples with 108 train-split masking samples. Since we perform no training on MELD, the train/test distinction is irrelevant to our experimental validity. The final dataset consists of 232 incongruent samples: 144 masking and 88 overstatement. Table 1 summarizes the sample distribution.

3.3 ESConv

For strategy-guided fine-tuning, we use ESConv (Liu et al., 2021), containing 1,053 conversations (18,376 utterances) between crowdworkers in supporter and help-seeker roles, each supporter utterance annotated with one of eight strategy labels

Split	Total	Mismatch	Masking	Overstatement
Train	9,989	441	108	333
Test	2,610	124	36	88
Used	—	232	144	88

Table 1: Mismatch sample distribution in MELD. We use all 88 overstatement samples from the test split; since we perform no training on MELD, we supplement only the masking samples (36 test-split) with 108 train-split samples to partially balance the two subtypes.

grounded in Helping Skills Theory (Hill, 2009). We exclude *Information* and *Others* as too domain-specific or underspecified, yielding six strategies for fine-tuning and inference. Each training sample is formatted as a complete chat sequence using Qwen2.5-7B-Instruct’s native chat template: a system prompt establishing the assistant role, a user turn containing the conversation history and a strategy instruction of the form “Please respond to the user using the following strategy: [strategy name]”, and an assistant turn containing the target supporter response. Strategy conditioning is thus achieved through natural language instruction rather than special tokens or architectural modification.

4 Method

4.1 Overview

Our approach proceeds in two stages: first establishing that mismatch awareness improves empathic response quality by replicating the core trend of Tahir et al. (2025) under a reproducible prompting-based setting, then asking the more fine-grained question of which response strategy works best for which incongruence type. The second stage treats strategy identity as the independent variable while controlling for model, prompt structure, and mismatch context, allowing us to empirically discover type-specific strategy affinities through LoRA fine-tuning on ESConv evaluated across both masking and overstatement samples.

4.2 Stage 1: Prompting-Based Baselines

Baseline reproduction. We reproduce the central empirical claim of Tahir et al. (2025)—that incongruence-aware conditioning improves empathic response quality—under a fully reproducible setting. Because E-THER’s dataset, fine-tuned weights, and proprietary keyword-based evaluators are not publicly released, we cannot run their exact pipeline. We therefore reproduce the *trend* rather than the *artifact*: we substitute (i) the model

(Qwen2.5-7B-Instruct (Team, 2025) in place of GPT-4V), (ii) the dataset (MELD’s incongruent subset in place of E-THER’s clinical videos), (iii) the conditioning mechanism (prompting in place of fine-tuning), and (iv) the evaluators (a pretrained empathy classifier and LLM judges in place of proprietary keyword banks). A successful reproduction is one in which adding mismatch information to the prompt monotonically increases empathy and acknowledgment scores, regardless of absolute scale. We chose Qwen2.5-7B-Instruct because it is open, strong on instruction following, and small enough to run all 1,624 generations and all judge calls on a single H100 within the project budget.

Conditions. We evaluate four prompting-based conditions using Qwen2.5-7B-Instruct:

Text-Only: the model receives only the utterance text and is prompted to respond empathetically, with no access to acoustic information.

Audio-Only: the model receives the utterance text and the audio emotion label, instructed to respond based on how the user sounds rather than what they said.

Naive Fusion: the model receives both the utterance text and audio emotion label with a standard empathic response instruction, without any explicit mention of incongruence between the two modalities.

Mismatch Acknowledgment: the model receives the utterance text, audio emotion label, and an explicit description of the incongruence and its subtype (masking or overstatement), instructed to gently acknowledge the mismatch before responding.

These four conditions form a progression from no acoustic awareness to full explicit incongruence acknowledgment, allowing us to isolate the contribution of each level of mismatch awareness. We evaluate all four conditions using three metrics—Empathy Score, Human Tone, and Acknowledgment Score—which correspond to the same underlying empathic constructs measured by E-THER’s proprietary evaluators: PCT Adherence (overall therapeutic empathy), Empathic Authenticity (naturalness and warmth), and Responsive Engagement (whether the model attends to the user’s specific emotional state). Since E-THER’s keyword banks are not publicly available, we implement these constructs using a pretrained empathy classifier and LLM judges, which have been shown to approximate human judgment more reliably than keyword

matching (Liu et al., 2016; Zheng et al., 2023). Full prompt templates are provided in Appendix A.

4.3 Stage 2: Strategy-Guided Fine-Tuning

Rationale. Stage 1 confirms that mismatch awareness helps, but treats every incongruence the same way—inheriting the central limitation of Tahir et al. (2025), which models incongruence as a single undifferentiated phenomenon. Yet masking (suppressed distress) and overstatement (sarcasm or playful exaggeration) carry opposite communicative intents, so a uniform empathic response is unlikely to serve both equally. Stage 2 therefore swaps undifferentiated mismatch-awareness for explicit *strategy* conditioning grounded in Hill’s Helping Skills theory (Hill, 2009), treating strategy choice as the independent variable and incongruence type as the moderator. This design lets us empirically test whether type-specific strategy affinities exist—a question the prior literature could not answer because no prior work pairs ESC strategy labels with acoustic-semantic mismatch.

We fine-tune Qwen2.5-7B-Instruct on ESConv (Liu et al., 2021) using LoRA (Hu et al., 2022), targeting the attention projection layers (q_proj, k_proj, v_proj, o_proj) with rank $r = 16$ and scaling factor $\alpha = 32$. Training runs for 3 epochs on 18,376 ESConv utterances with a learning rate of 2×10^{-4} and cosine scheduling.

At inference time, the fine-tuned model is prompted with the mismatch context and conditioned on one of six ESConv strategies: *Question*, *Reflection of Feelings*, *Affirmation and Reassurance*, *Restatement or Paraphrasing*, *Providing Suggestions*, and *Self-Disclosure*. Each of the 232 incongruent samples receives one response per strategy, plus one No-Strategy baseline response (introduced below), yielding 1,624 total responses for evaluation (232×7 conditions).

Within-Experiment Baseline. A key methodological challenge is that fine-tuning on ESConv changes model behavior—in particular, it reduces the model’s tendency to explicitly acknowledge mismatches, making the external prompting baseline no longer directly comparable to the fine-tuned model. We therefore introduce a within-experiment baseline (*No Strategy*), where the same fine-tuned model receives the same mismatch-aware prompt but without any strategy conditioning, establishing a lower bound for strategy-guided generation. All comparisons are controlled for model identity and

prompt structure, isolating the effect of deliberate strategy selection alone.

4.4 Evaluation Metrics

Since E-THER’s evaluation metrics rely on proprietary keyword banks that are not publicly available, direct adoption is not feasible. Furthermore, keyword-matching metrics are known to correlate poorly with human judgment in dialogue evaluation (Liu et al., 2016). We therefore design an evaluation suite that targets the same underlying empathic constructs as E-THER, implemented with more expressive and reproducible methods.

Empathy Score applies a RoBERTa-based empathy classifier (Loh, 2022) to each generated response, producing a continuous score of overall empathic quality. This corresponds to E-THER’s PCT Adherence metric, which aggregates Rogers’ core therapeutic conditions.

Human Tone (1–5) uses Qwen2.5-7B-Instruct as an LLM judge to rate response naturalness and warmth, corresponding to E-THER’s Empathic Authenticity metric. The judge is prompted to distinguish genuine conversational engagement from performative or robotic language, scoring on a 1–5 scale where 1 is robotic and cold and 5 is very natural and warm.

Acknowledgment Score (0/1) uses an LLM judge to determine whether the response explicitly contrasts the user’s words against their vocal tone, corresponding to E-THER’s Responsive Engagement metric, operationalized as whether the mismatch itself is surfaced in the response.

Emotion Correctness and **Appropriateness** are two additional metrics introduced for the strategy fine-tuning stage. Emotion Correctness assesses whether the response correctly identifies the user’s underlying emotional state given the incongruence type: for masking, this means recognizing suppressed negative distress; for overstatement, recognizing that expressed negative language overstates actual distress. Appropriateness assesses whether the chosen response strategy fits the specific incongruence type and conversational context, penalizing responses that apply a mismatched strategy. Both metrics are evaluated at two scales: a binary (0/1) judgment for coarse-grained comparison across strategies, and a graded (1–5) score to capture finer distinctions that binary labels may obscure—for instance, distinguishing a partially appropriate response from one that is entirely off-target. Reporting both scales allows us to verify

whether the two levels of granularity yield consistent conclusions.

Together, these five metrics cover empathic quality, naturalness, incongruence awareness, emotional understanding, and strategy fit, providing a comprehensive view of response quality across both incongruence types.

4.5 Implementation Details

All experiments use Qwen2.5-7B-Instruct as the base model. LoRA fine-tuning is conducted on NVIDIA H100 GPUs with BF16 precision, batch size of 2, gradient accumulation over 8 steps (effective batch size 16), learning rate of 2×10^{-4} with cosine scheduling, and 100 warmup steps. The LLM judge uses the same model loaded in FP16 with greedy decoding (do_sample=False) and a maximum of 8 new tokens to ensure deterministic and concise judgments.

5 Results

5.1 Baseline Results

Table 2 shows results across the four prompting-based baseline conditions. Empathy score and acknowledgment rate increase monotonically as mismatch information is progressively introduced, with Mismatch Acknowledgment achieving the highest empathy (0.032) and acknowledgment rate (0.737), aligning with E-THER’s central finding that incongruence awareness improves empathic response quality. Text-Only scores highest on Human Tone (4.871), with naturalness declining as more task constraints are added—a trade-off we attribute to the difference between explicit prompting and implicit fine-tuning: instructing the model to acknowledge a mismatch introduces a task-oriented tone that reduces conversational fluency. Audio-Only performs worst on Human Tone (3.948), suggesting that acoustic information without textual grounding disrupts response naturalness.

Condition	Empathy	Tone	Ack
Text-Only	0.0178	4.871	0.224
Audio-Only	0.0221	3.948	0.319
Naïve Fusion	0.0207	4.522	0.293
Mismatch Ack	0.0318	4.435	0.737

Table 2: Average scores across four prompting-based baseline conditions on 232 mismatch samples. Bold indicates best per metric.

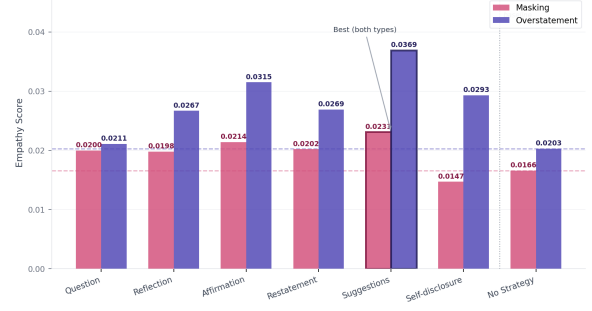


Figure 1: Empathy Score by strategy for masking and overstatement samples. Overstatement consistently outperforms masking across all strategies, with the largest gap observed for Providing Suggestions (0.0369 vs. 0.0231).

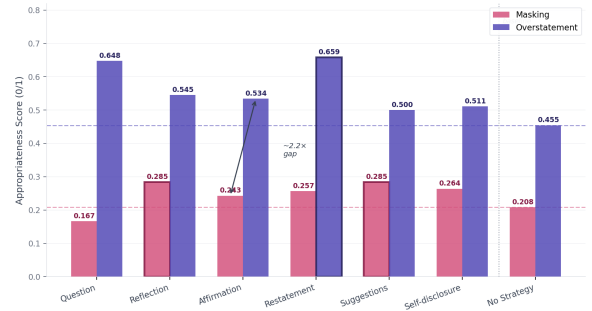


Figure 2: Appropriateness Score (Ap 1–5) by strategy for masking and overstatement samples. Overstatement scores 2–3 $\times$ higher than masking across all strategies. Restatement or Paraphrasing achieves the highest appropriateness for overstatement but not for masking, demonstrating that the two incongruence types require different response strategies.

5.2 Strategy Fine-Tuning Results

Given the relatively small sample sizes ($n=144$ masking, $n=88$ overstatement) and the use of LLM-based evaluators, we treat results as indicative trends rather than definitive claims and do not report significance tests. Tables 3 and 4 report per-strategy results for masking and overstatement respectively; Figures 1 and 2 visualize Empathy and Appropriateness scores across both incongruence types.

5.3 Analysis

Strategy outperforms no-strategy baseline. All six strategies achieve higher empathy than No Strategy on overstatement (0.0211–0.0369 vs. 0.0203), and five of six on masking (0.0147–0.0231 vs. 0.0166), with Self-Disclosure the sole exception (0.0147). Strategy gains are also visible on Tone, Ack, EC, and Ap (Tables 3, 4), confirming that

Strategy	Empathy	Tone	Ack	EC(0/1)	EC(1-5)	Ap(0/1)	Ap(1-5)
Question	0.0200	2.993	0.063	0.007	1.979	0.167	2.278
Reflection of Feelings	0.0198	2.563	0.063	0.111	2.111	0.285	2.285
Affirmation & Reassurance	0.0214	2.951	0.021	0.042	1.924	0.243	2.257
Restatement or Paraphrasing	0.0202	2.750	0.028	0.056	2.035	0.257	2.333
Providing Suggestions	0.0231	3.111	0.056	0.028	2.063	0.285	2.424
Self-Disclosure	0.0147	2.639	0.028	0.014	2.076	0.264	2.319
No Strategy	0.0166	2.910	0.042	0.021	1.833	0.208	2.146

Table 3: Results on masking samples (n=144). EC = Emotion Correctness, Ap = Appropriateness, reported at both binary (0/1) and graded (1–5) scales. Bold indicates best performance per metric.

Strategy	Empathy	Tone	Ack	EC(0/1)	EC(1-5)	Ap(0/1)	Ap(1-5)
Question	0.0211	3.102	0.068	0.011	2.261	0.648	2.807
Reflection of Feelings	0.0267	2.989	0.080	0.080	2.466	0.545	2.705
Affirmation & Reassurance	0.0315	3.250	0.034	0.045	2.409	0.534	2.761
Restatement or Paraphrasing	0.0269	3.239	0.045	0.057	2.375	0.659	2.807
Providing Suggestions	0.0369	3.057	0.068	0.011	2.386	0.500	2.830
Self-Disclosure	0.0293	3.011	0.023	0.023	2.409	0.511	2.818
No Strategy	0.0203	3.102	0.045	0.034	2.273	0.455	2.818

Table 4: Results on overstatement samples (n=88). EC = Emotion Correctness, Ap = Appropriateness, reported at both binary (0/1) and graded (1–5) scales. Bold indicates best performance per metric.

explicit strategy conditioning provides consistent improvements over unguided generation under the same fine-tuned model and prompt structure.

Providing Suggestions is the most robust strategy overall. Providing Suggestions achieves the highest empathy on both masking (0.0231) and overstatement (0.0369), the highest Human Tone on masking (3.111), and the highest Appropriateness on overstatement (2.830), suggesting that action-oriented, forward-looking responses are broadly effective by redirecting attention away from the mismatch toward constructive engagement.

Different strategies show divergent performance across incongruence types. Strategy effectiveness is not consistent across mismatch subtypes. Restatement or Paraphrasing achieves the highest binary Appropriateness on overstatement (0.659) but ranks among the lower strategies on masking (0.257), a gap of over 0.4 points. Conversely, Reflection of Feelings achieves the highest Emotion Correctness on masking (EC 0/1: 0.111, EC 1-5: 2.111)—the only strategy to meaningfully recognize suppressed distress—but performs more modestly on overstatement, confirming that the two subtypes require qualitatively different response strategies.

Overstatement is consistently easier than masking. Overstatement samples outperform masking across all strategies and nearly all metrics: Appropriateness scores are 2–3 $\times$ higher (0.50–0.66 vs. 0.17–0.29) and empathy scores are uniformly higher. We attribute this asymmetry to the nature of the two subtypes—overstatement involves exaggerated language that is relatively transparent in context, while masking involves actively suppressed emotion that is harder for the model to identify and address appropriately.

Fine-tuning trades off acknowledgment for strategy adherence. Ack scores remain very low across all strategy conditions (0.02–0.08), substantially below the Mismatch Acknowledgment baseline (0.737). ESConv fine-tuning instills strong strategy adherence but suppresses the tendency to explicitly surface the incongruence—a trade-off suggesting that future work should incorporate mismatch-aware training data to recover acknowledgment ability alongside strategy guidance.

5.4 Qualitative Analysis

Inspection of generated responses (Appendix B) confirms the type-specific strategy patterns. For masking, *Reflection of Feelings* and *Providing Suggestions* successfully infer suppressed negative emotion from positive surface framing and redirect toward constructive engagement. For overstate-

ment, *Restatement or Paraphrasing* normalises exaggerated expression by mirroring it calmly without amplification—the same mirroring move applied to masking risks reinforcing the surface positive affect instead of probing beneath it. These patterns confirm that the central challenge for masking is *depth of inference* (detecting what is not said) while for overstatement it is *calibration* (neither dismissing nor amplifying).

5.5 Error Analysis

Manual inspection (Appendix C) surfaces three recurring failure patterns common to baseline and fine-tuned conditions. **(1) Context-free short utterances** (e.g., “Push!”, “Okay!”) have no recoverable semantic content in isolation, so all conditions produce generic responses—a structural artefact of scripted TV dialogue that depresses metrics uniformly. **(2) Non-self-expressive mislabelling** arises when our polarity-based pipeline tags jokes, narration, or compliments to others as masking (e.g., “I’ve never seen either of you one/millionth as happy...”), introducing label noise inherent to general-purpose sentiment classification on multi-speaker dialogue. **(3) Sarcastic masking** (e.g., “Oh, this is just perfect!”) is lexically positive but pragmatically negative; neither baseline nor fine-tuned models reliably handle it, as reflected by uniformly low Emotion Correctness (EC 0/1: 0.007–0.111). Binary polarity comparison is insufficient to capture sarcasm and requires dedicated modeling.

6 Additional Experiment: Audio Language Model Fine-Tuning

Stage 2 reveals a systematic trade-off: ESConv fine-tuning improves strategy adherence but reduces mismatch acknowledgment from 0.737 to 0.02–0.08. A natural follow-up is whether replacing the audio-emotion-label intermediate with an *end-to-end* audio language model (ALM) can recover acknowledgment without sacrificing naturalness. We test this with Qwen2-Audio-7B-Instruct (Chu et al., 2024), an open ALM that ingests raw 16 kHz audio alongside the transcript, evaluated on the same 232 incongruent samples and metrics as Stage 2.

Setup. We fine-tune Qwen2-Audio-7B-Instruct with QLoRA (rank $r=16$, $\alpha=32$, dropout 0.05, attention projections only) on a 982-sample mismatch-acknowledgment corpus comprising **441**

real MELD-train incongruent utterances paired with curated mismatch-aware target responses, plus **541 synthetic** examples generated via few-shot prompting on ESConv-style contexts. Training runs for 3 epochs at learning rate 2×10^{-4} on a single H100 with 4-bit base-model quantization. We compare four conditions: **ALM-TextAudio** (transcript + audio, generic empathic instruction), **ALM-AudioOnly** (audio waveform alone), **ALM-MismatchAck** (text + audio with the Stage 1 mismatch-aware prompt), and **ALM-Finetuned** (the QLoRA-fine-tuned model with the mismatch-aware system prompt used during training).

Condition	Empathy	Tone	Ack
ALM-TextAudio	0.018	4.05	0.211
ALM-AudioOnly	0.016	3.39	0.285
ALM-MismatchAck	0.027	4.05	0.647
ALM-Finetuned (ours)	0.046	4.29	0.690

Table 5: ALM evaluation on the same 232 mismatch samples (Qwen2-Audio-7B-Instruct). Bold indicates best per metric.

ALM fine-tuning recovers the acknowledgment lost in Stage 2. Replacing the emotion-label intermediate with raw audio raises Ack from 0.211 (ALM-TextAudio) to 0.647 under matched prompting (ALM-MismatchAck), and QLoRA fine-tuning further pushes it to 0.690—a **3.3 $\times$** improvement over the ALM baseline. Crucially, this is achieved *without* sacrificing naturalness: tone rises from 4.05 to 4.29, and empathy more than doubles (0.018 $\rightarrow$ 0.046). Compared with the Stage 2 fine-tuned LLM (Ack 0.02–0.08), the ALM closes nearly the entire acknowledgment gap that ESConv adaptation opened up. The result implies that end-to-end audio access—not just emotion-label conditioning—is what enables consistent mismatch surfacing, and that strategy adherence (from ESConv-style fine-tuning) and mismatch awareness (from ALM fine-tuning) are complementary capabilities that future ESC systems should fuse rather than choose between.

7 Discussion

Our results confirm that strategy-guided fine-tuning outperforms the prompting-based No Strategy baseline across both incongruence types, demonstrating that ESConv fine-tuning successfully transfers strategy knowledge to mismatch scenarios even without explicit incongruence supervision. We in-

interpret these findings in terms of problem structure, method strengths and limitations, and broader implications for multimodal ESC system design.

Why Providing Suggestions is broadly effective.

The consistent superiority of Providing Suggestions across both incongruence types suggests that action-oriented responses offer a strategic advantage in mismatch scenarios: rather than requiring precise judgment about the user’s underlying emotional state—a task complicated by the very nature of incongruence—they redirect the conversation toward constructive engagement, sidestepping the uncertainty introduced by the mismatch. In the absence of fine-grained incongruence understanding, action-oriented strategies thus serve as a safe and effective default for multimodal ESC systems.

Why masking is harder than overstatement.

The systematic performance gap across all metrics points to a fundamental asymmetry in inferential difficulty. Overstatement presents a tractable mapping: exaggerated negative language and positive vocal tone together signal that expressed distress overstates actual emotional state, a pattern transparent enough for both model and evaluators. Masking requires inferring an emotional state the speaker is actively concealing—the surface signal is positive, and the true state must be reconstructed from the acoustic-text conflict alone. This higher-order inferential task exceeds current LLMs’ reliable capabilities even after empathic fine-tuning, as confirmed by uniformly low Emotion Correctness scores (EC 0/1: 0.007–0.111), suggesting masking represents an underexplored and genuinely difficult frontier for multimodal ESC.

The acknowledgment–strategy trade-off. The dramatic Ack score drop after fine-tuning (0.737 → 0.02–0.08) reveals a fundamental tension between explicit mismatch acknowledgment and strategy-guided generation. ESConv fine-tuning instills strong strategy adherence but suppresses incongruence surfacing, because ESConv dialogues provide no training signal for acknowledgment behavior. The two capabilities cannot be acquired simultaneously from a single corpus; future work must either construct mismatch-aware training data combining strategy labels with incongruence annotations, or adopt multi-objective training that explicitly balances the two.

Emotion correctness and appropriateness capture different dimensions. High Appropriateness

does not necessarily co-occur with high Emotion Correctness: Restatement or Paraphrasing leads on binary Appropriateness for overstatement (0.659) but not Emotion Correctness, while Reflection of Feelings leads on Emotion Correctness for masking (0.111) but not Appropriateness. This dissociation confirms that strategy fit and emotional understanding are distinct dimensions of empathic quality—a response can adopt a contextually appropriate strategy while misidentifying the user’s emotional state, or vice versa—and that effective ESC systems must optimize both simultaneously. Our five-metric suite is designed precisely to surface these distinctions.

Implications for multimodal ESC system design.

Our findings argue against the assumption that a single empathic strategy can serve all users equally well. When acoustic and semantic signals conflict, the appropriate response depends critically on incongruence type: overstatement calls for normalization and reframing, masking for gentle probing of suppressed distress—qualitatively different communicative acts requiring different model capabilities. Practically, multimodal ESC systems should incorporate an incongruence-type classifier as an upstream component routing inputs to type-specific strategy modules. Theoretically, the field needs richer incongruence taxonomies beyond the binary masking/overstatement distinction to capture subtypes such as sarcasm, irony, and context-dependent suppression, each requiring further specialized strategies.

8 Conclusion and Future Work

Recap. We investigated response strategy selection under acoustic-semantic emotional incongruence in supportive dialogue, asking whether *masking* and *overstatement* require different empathic response strategies. Through a two-stage evaluation—four prompting-based baselines that reproduce the mismatch-awareness trend of [Tahir et al. \(2025\)](#), followed by strategy-guided LoRA fine-tuning on ESConv—we find that explicit mismatch acknowledgment substantially raises empathy and acknowledgment (0.018 → 0.032 and 0.22 → 0.74), that *Providing Suggestions* is the most robust strategy across both subtypes, and that strategy effectiveness genuinely depends on incongruence type (e.g., *Restatement or Paraphrasing* on overstatement Ap 0.659 vs. on masking 0.257). Overstatement is consistently more tractable than masking, and fine-

tuning trades off mismatch acknowledgment for strategy adherence.

Broader takeaway. Beyond these specific findings, our results show that incongruence type is a meaningful axis of variation that uniform-strategy and text-only ESC systems are not equipped to handle. This argues for a shift toward type-sensitive, modality-aware ESC design, in which an upstream incongruence-type classifier routes inputs to type-specific strategy modules rather than relying on a single empathic default. The reproducible artifacts we contribute—a controlled within-experiment baseline design and a five-metric evaluation suite covering empathy, naturalness, acknowledgment, emotion correctness, and strategy fit—are independent of any particular model or dataset and should transfer to future multimodal ESC research.

Future work should pursue three directions. First, constructing mismatch-aware training data that combines ESConv-style strategy annotations with incongruence labels would allow models to learn both strategy adherence and mismatch acknowledgment simultaneously. Second, extending the incongruence taxonomy beyond the binary masking/overstatement distinction to include subtypes such as sarcasm, irony, and context-dependent suppression would enable finer-grained strategy matching. Third, scaling to larger models and more naturalistic datasets beyond scripted television dialogue would test whether the type-strategy affinities we identify generalize to real-world emotional support interactions.

9 Limitations

Several limitations should be considered when interpreting our findings. First, our mismatch samples are drawn exclusively from MELD, a dataset of scripted television dialogue from *Friends*, which may not represent the full range of naturalistic emotional expression. The scripted nature of the source material introduces artifacts such as performative emotion and context-dependent utterances that are difficult to interpret in isolation. Second, our mismatch detection pipeline relies on binary sentiment polarity and does not capture pragmatic phenomena such as sarcasm, irony, or culturally specific emotional expression, leading to label noise that affects all experimental conditions. Third, all five evaluation metrics are automatic and none have been validated against human judgments in our specific setting; LLM judges in particular may exhibit bi-

ases toward certain response styles regardless of true empathic quality. Fourth, our experiments use a single base model (Qwen2.5-7B-Instruct) and a single fine-tuning dataset (ESConv), limiting the generalizability of our strategy-type affinity findings to other model families and training corpora.

10 Ethical Considerations

This work raises several ethical considerations. First, MELD is derived from a commercial television series and contains portrayals of emotion that reflect the demographics and cultural norms of its cast and writers; findings based on this data may not generalize to underrepresented populations or non-Western emotional expression styles. Second, automated empathic response systems carry inherent risks when deployed in real emotional support contexts: a model that misidentifies a user’s emotional state—as our results show current systems frequently do for masking cases—may produce responses that feel dismissive or inappropriate, potentially causing harm to vulnerable users. Third, the use of an LLM as both the response generator and the evaluator (Human Tone, Acknowledgment, Emotion Correctness, Appropriateness) introduces a circularity risk, as the judge model may favor responses that resemble its own generation style. Researchers building on this work should validate LLM judge assessments against human annotations before deployment. Finally, any system trained to detect emotional incongruence could in principle be misused for deceptive purposes—for instance, identifying when individuals are concealing distress in contexts where such detection is not consented to. We encourage researchers and practitioners to apply such systems only in contexts where users are informed and consent to multimodal emotional analysis.

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A Prompt Templates

All four baseline conditions use Qwen2.5-7B-Instruct with greedy decoding (do_sample=False, max_new_tokens=150). Prompts are formatted using the model's built-in chat template. The placeholder [utterance] refers to the transcribed user utterance and [audio_emotion] refers to the MELD audio emotion label.

Text-Only

A user said: "[utterance]"
Respond empathetically as an emotional support assistant.

Audio-Only

A user said: "[utterance]"
The user's voice sounds [audio_emotion].
Respond based on how the user sounds, not just what they said.

Naive Fusion

A user said: "[utterance]"
The user's voice sounds [audio_emotion].
Respond empathetically as an emotional support assistant.

Mismatch Acknowledgment — Masking

Applied when text sentiment is positive and audio sentiment is negative.

A user said: "[utterance]"
The user's voice sounds [audio_emotion].
There is an emotional mismatch: the user's words seem neutral or positive, but their voice sounds [audio_emotion], suggesting they may be hiding or downplaying their true feelings.
Acknowledge this mismatch gently and respond empathetically as an emotional support assistant.

Mismatch Acknowledgment — Overstatement

Applied when text sentiment is negative and audio sentiment is positive.

A user said: "[utterance]"
The user's voice sounds [audio_emotion].

There is an emotional mismatch: the user's words seem more intense or negative, but their voice sounds [audio_emotion], suggesting they may be exaggerating or overstating their feelings.
Acknowledge this mismatch gently and respond empathetically as an emotional support assistant.

B Qualitative Analysis Examples

Masking Example 1: Reflection of Feelings (ID 96)

- **Utterance:** “I mean, enough of the silliness!”
- **Type:** Masking (text=joy, audio=anger)
- **Reflection of Feelings response:** “It sounds like you’re frustrated with your friends for not being supportive during this difficult time.”
- **Analysis:** Successfully infers suppressed negative emotion despite surface-level positive framing. The response correctly identifies underlying frustration rather than taking the utterance at face value.

Masking Example 2: Providing Suggestions (ID 2299)

- **Utterance:** “Oh... Mike’s sister just invited me to a party tonight, he’s gonna be there.”
- **Type:** Masking (text=joy, audio=anger)
- **Providing Suggestions response:** “I see. I’m sorry you’re angry about this. Do you think you should go to the party anyway?”
- **Analysis:** Recognizes suppressed anger and redirects toward constructive action, demonstrating that action-oriented strategies can effectively handle masking by acknowledging the hidden emotion while providing forward-looking guidance.

Overstatement Example: Restatement or Paraphrasing (ID 273)

- **Utterance:** “No, but I want... I want the pinecones!”
- **Type:** Overstatement (text=anger, audio=joy)
- **Restatement or Paraphrasing response:** “So you want the pine cones?”
- **Analysis:** Mirrors back the utterance in a calm, normalized form without amplifying the exaggerated language. Contrast with masking: applying the same mirroring strategy to a masking case would risk reinforcing the surface positive affect rather than probing the suppressed distress beneath.

C Error Analysis Examples

Pattern 1: Context-Free Short Utterances (ID 134)

- **Utterance:** “Okay!”
- **Type:** Overstatement (text=anger, audio=joy)
- **Text-Only response:** “I’m here for you, and it sounds like you’re okay for now...”
- **Mismatch Ack response:** “It sounds like you might be using ‘Okay!’ to express something other than just agreement...”
- **Empathy:** Text-Only 0.016 → Mismatch Ack 0.021
- **Issue:** Single-word utterance provides no semantic content. All strategies produce generic responses regardless of mismatch information.

Pattern 2: Non-Self-Expressive Utterance (ID 99)

- **Utterance:** “I’ve never seen either of you one/millionth as happy as you’ve been since you’ve got together.”
- **Type:** Masking (text=joy, audio=sadness)
- **Mismatch Ack response:** “It sounds like you might be feeling a bit sad about something else...”
- **Empathy:** 0.012, Ack: 0
- **Issue:** Speaker is complimenting others, not self-disclosing. Model incorrectly applies masking interpretation, producing an inappropriate response and failing to acknowledge any mismatch.

Pattern 3: Sarcastic Masking Not Recognized (ID 913)

- **Utterance:** “Ohh, well, this is just perfect!”
- **Type:** Masking (text=joy, audio=anger)
- **Mismatch Ack response:** “Is everything really perfect, or is there something bothering you?”
- **Ack Score:** 0

- **Issue:** Despite being informed of the mismatch, the model fails to acknowledge it. The sarcastic phrasing “just perfect” is not captured by the masking prompt, which assumes genuinely positive words rather than sarcasm.